



COMPETENCY STANDARD

FOR

BATCHING, PREPARING AND DRAWING

(JUTE SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for **Batching, Preparing and Drawing** is designed to guide the development of curricula, teaching materials, and assessment tools. It also serves as a framework for providing industry-aligned training, ensuring that individuals meeting the prescribed standards through assessments are qualified to secure relevant employment opportunities.

This document was originally developed under the **Skills for Employment Investment Program (SEIP)** and later reviewed and updated with input from occupation-specific industry experts to be utilized in training initiatives under the **Skills for Industry Competitiveness and Innovation Program (SICIP)**. The document is owned by the Finance Division, Ministry of Finance, People's Republic of Bangladesh.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national and international process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:**Generic Competencies (20 Hrs.)**

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-JUTE-BPD-01-G	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Apply personal health and safety practices 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-JUTE-BPD-02-G	Work in a team environment	1. Identify team goals and processes 2. Communicate and cooperate with team members 3. Work as a team member 4. Solve problems as a team member	10
Total Hour			20 Hrs.

Sector Specific Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-JUTE-BPD-01-S	Work in the Jute Sector	1. Identify the jute sector and its main activities 2. Understand basic roles in jute production processes 3. Recognize common tools and equipment used 4. Understand simple quality requirements in jute work	10
SICIP-JUTE-BPD-02-S	Apply Green Practices in Jute Sector	1. Interpret green concepts 2. Minimize resources use in the workplace 3. Implement waste management practices	10
Total Hour			20 Hrs.

Occupation Specific Competencies (180 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-JUTE-BPD-01-O	Interpret Materials, Equipment and Processes	1. Identify raw jute and grades 2. Identify batching oils and chemicals 3. Identify tools and processing machine 4. Identify processing machine parts and functions 5. Describe batching and preparing methods 6. Describe drawing process	20
SICIP-JUTE-BPD-02-O	Perform Batching Operation	1. Select and sort out jute for batching 2. Measure oil, water and emulsifier ratios 3. Prepare emulsion solutions 4. Operate softener and spreader machine 5. Store piled jute properly	70
SICIP-JUTE-BPD-03-O	Perform Jute Preparing Operations	1. Feed piled jute into preparing machines 2. Adjust machine settings and controls 3. Operate breaker card and finisher card machine 4. Monitor fibre condition and output 5. Maintain continuous material flow	40
SICIP-JUTE-BPD-04-O	Perform Drawing Operations	1. Feed slivers into drawing machines 2. Adjust draft and speed settings 3. Operate drawing frames 4. Monitor sliver quality and uniformity 5. Handle drawing output materials properly	30
SICIP-JUTE-BPD-05-O	Maintain Equipment and Work Area	1. Perform cleaning of machines and work areas 2. Conduct routine equipment checking 3. Carry out basic machine maintenance	20

Code	Unit of Competency	Elements of Competency	Duration (hours)
		4. Maintain and store tools and equipment properly	
SICIP-JUTE-BPD-06-O	Maintain Quality and Production Efficiency	1. Check product quality standards 2. Identify defects and irregularities 3. Take corrective actions during production 4. Maintain production targets and efficiency 5. Follow workplace rules and procedures	20
Total Hours			0 Hrs.

COMPETENCY STANDARD: BATCHING, PREPARING AND DRAWING

Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-JUTE-BPD-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply Occupational Health and Safety (OH&S) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures; applying personal health and safety practices; reporting hazards and risks; and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Apply personal health and safety practices	2.1 OHS policies and procedures are followed and practiced. 2.2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2.3 Personal hygiene is maintained.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 <u>Emergency response plans and procedures</u> are implemented. 4.3 <u>First aid procedure</u> is applied during emergency situations.

Range of Variables

Variable	Range
	May include but not limited to:
1. OHS policies	1.1 International OHS requirements 1.2 Bangladesh standards for OHS 1.3 Building Code

	1.4 Fire Safety Rules and Regulations 1.5 Industry Guidelines
2. Personal Protective Equipment (PPE)	2.1 Apron 2.2 Gas Mask 2.3 Gloves 2.4 Safety shoes 2.5 Helmet 2.6 Face mask 2.7 Overalls 2.8 Goggles and safety glasses 2.9 Ear plugs 2.10 Sun block 2.11 Chemical/Gas masks
3. Hazards and risks	3.1 Chemical hazards. 3.2 Biological hazards. 3.3 Physical Hazards. 3.3.1 Machine hazards. 3.3.2 Materials hazards. 3.3.3 Tools and Equipment hazards.
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first aid
5. First aid procedure	5.1 Washing of open wound 5.2 Washing chemically infected area 5.3 Applying bandage 5.4 Tourniquet 5.5 Applying CPR (Cardiopulmonary Resuscitation) 5.6 Taking appropriate medicine

Curricular Evidence Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures. 1.3 Emergency procedures. 1.3.1 Firefighting. 1.3.2 Earthquake response. 1.3.3 Explosion response. 1.3.4 Accident response. 1.4 Types of (biological, chemical and physical) and their effects. 1.5 PPE types and uses. 1.6 Personal hygiene practices. 1.7 OHS awareness.
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Following personal work safety practices 2.3 Reporting hazards and risks

	2.4 Responding to emergency procedures 2.5 Maintaining physical well-being in the workplace 2.6 Performing first aid. 2.7 Performing basic firefighting accessories using fire extinguishers 2.8 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Communication with peers, sub-ordinates and seniors in workplace. 3.3 Promptness in carrying out activities 3.4 Tidiness and timeliness 3.5 Respect of peers, sub-ordinates and seniors in workplace 3.6 Environmental concern 3.7 Sincere and honest to duties
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Followed OHS policies and procedures. 1.2 Selected and used personal protective equipment (PPE). 1.3 Reported incidents arising from hazards and risks to authority. 1.4 Emergency response plans and procedures are implemented. 1.5 Applied basic first aid procedure.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: WORK IN A TEAM ENVIRONMENT	Nominal Duration: 10 hrs.	Unit Code: SICIP-JUTE-BPD-02-G
Unit Descriptor:		

This unit covers the skills, knowledge and attitudes required to work in a team environment. It specifically includes the tasks of identifying team goals and processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within team and with other workers are identified.
2. Communicate and cooperate with team members	2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2.2 Formal and informal <u>forms of communication</u> are used effectively to support team achievement. 2.3 Diversity in character is respected and valued in team functioning. 2.4 Views and opinions of other team members are understood and valued. 2.5 Workplace terminology is used correctly to assist communication.
3. Work as a team member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team. 3.2 Tasks are performed in accordance with organizational and team requirements, specifications, and workplace procedures. 3.3 Team member's support with other members is made to ensure team achieves goals, awareness, and requirements. 3.4 Agreed reporting lines are followed using standard operating procedure.
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of variables:

Variables	Range (may include but not limited to)
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1. Forms of communication	1.1	Agenda
	1.2	Simple reports such as progress and incident reports
	1.3	Job sheets
	1.4	Operational manuals
	1.5	Brochures and promotional material
	1.6	Visual and graphic materials
	1.7	Standards
	1.8	OSH information
	1.9	Signs

Curricular Content Guide

1. Underpinning Knowledge	1.1	Team goals and collaborative decision-making processes
	1.2	Roles and responsibilities of team members
	1.3	Relationships within team and with other workers
	1.4	Effective interpersonal skills to interact with team members
	1.5	Effective formal and informal forms of communication
	1.6	Value of diversity in team functioning.
	1.7	Correct use of workplace terminology
	1.8	Team's duties, responsibilities, authorities, objectives, and task requirements
	1.9	Support mechanism to other members of team to ensure achievements of goals.
	1.10	Methods of identifying current and potential problems faced by a team
	1.11	Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2.1	Identifying team goals and collaborative decision-making processes
	2.2	Identifying roles and responsibilities of team members
	2.3	Identifying relationships within team and with other workers
	2.4	Using effective interpersonal skills to interact with team members and to contribute to activities and objectives
	2.5	Using formal and informal forms of communication
	2.6	Understanding and valuing views and opinions of other team members
	2.7	Performing tasks in accordance with organizational and team requirements, specifications, and workplace procedures.
	2.8	Supporting other members of the team to ensure team achieves goals, awareness, and requirements.
	2.9	Identifying current and potential problems faced by the team
	2.10	Identifying solutions to the problem
	2.11	Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1	Commitment to occupational health and safety
	3.2	Promptness in carrying out activities
	3.3	Sincere and honest to duties

	3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates, and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Manuals/Guidelines 4.3 Pens 4.4 Papers 4.5 Workbooks

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified team goals and work processes 1.2 communicated and cooperated with team members. 1.3 worked as a team member 1.4 solved problems as a team member
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: WORK IN THE JUTE SECTOR	Nominal Duration: 10 hrs.	Unit Code: SICIP-JUTE-BPD-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for work in the jute sector. It specifically includes the tasks of identifying the jute sector and its main activities, understanding basic roles in jute production processes, recognizing common tools and equipment used and understanding simple quality requirements in jute work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify the jute sector and its main activities	1.1 The jute sector is identified within the context of national and global industries. 1.2 <u>Main activities</u> are recognized. 1.3 <u>Different stages of jute value chain</u> are described in simple terms. 1.4 Key workplaces in the jute sector are identified. 1.5 Importance of the jute sector in the economy and environment is understood.
2. Understand basic roles in jute production processes	2.1 <u>Basic job roles in jute production</u> are identified and described. 2.2 Simple production workflows and task sequences are understood. 2.3 Teamwork and coordination in production processes are identified. 2.4 Workplace safety roles and responsibilities are understood and followed.
3. Recognize common tools and equipment used	3.1 <u>Common tools used in jute processing</u> are identified correctly. 3.2 <u>Basic machines and equipment used in production</u> are recognized. 3.3 Functions of tools and equipment are understood in simple terms. 3.4 Basic maintenance needs of tools and equipment are recognized.
4. Understand simple quality requirements in jute work	4.1 Basic quality standards in jute work are identified. 4.2 <u>Simple defects in jute products</u> are recognized. 4.3 Importance of maintaining quality during production is understood. 4.4 Work output is checked against basic quality expectations. 4.5 Good workmanship practices are followed to ensure

	quality.
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Range of Variables

Variable	Range May include but not limited to:
1. Main activities	1.1 Cultivation and harvesting of jute 1.2 Retting and fibre extraction 1.3 Baling and grading 1.4 Spinning and yarn production 1.5 Weaving and fabric production 1.6 Dyeing, printing, and finishing 1.7 Packaging and export
2. Different stages of jute value chain	2.1 Raw jute production (farm level) 2.2 Fibre processing (retting, stripping, drying) 2.3 Baling and grading 2.4 Spinning into yarn 2.5 Weaving/knitting into fabric 2.6 Finishing and value addition 2.7 Distribution and export
3. Basic job roles in jute production	3.1 Raw jute handler 3.2 Machine operator 3.3 Quality checker/inspector 3.4 Dyeing and finishing worker 3.5 Packing and store assistant 3.6 Supervisor or line in-charge
4. Common tools used in jute processing	4.1 Scissors and cutting knives 4.2 Measuring tape and rulers 4.3 Hand hooks and tongs 4.4 Weighing scale 4.5 Hand brushes
5. Basic machines and equipment used in production	5.1 Carding machine 5.2 Drawing machine 5.3 Spinning machine 5.4 Looms (handloom/power loom) 5.5 Dyeing vats or tanks 5.6 Baling press machine
6. Simple defects in jute products	6.1 Uneven thickness or slub in yarn 6.2 Broken fibres or yarn ends 6.3 Colour variation or patchiness 6.4 Holes or tears in fabric 6.5 Mis weave or weaving faults

Curricular Content Guide

1. Underpinning Knowledge	1.1 Overview of the jute sector within national and global industry context. 1.2 Main activities and stages of the jute value chain from
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	raw material to finished products. 1.3 Basic job roles, responsibilities, and workflows in jute production processes. 1.4 Importance of teamwork, coordination, and workplace safety in production settings. 1.5 Identification and functions of common tools, machines, and equipment used in jute work. 1.6 Basic maintenance awareness for tools and equipment used in production. 1.7 Understanding of basic quality standards, product defects, and quality expectations. 1.8 Principles of good workmanship and maintaining quality during jute production processes.
2. Underpinning Skills	2.1 Understanding basic job roles and responsibilities in jute production processes 2.2 Following simple production workflows, task sequences, and teamwork practices 2.3 Applying workplace safety roles and responsibilities in production activities 2.4 Understanding basic functions and maintenance needs of tools and equipment 2.5 Applying quality awareness during production to meet expected standards 2.6 Checking work output against basic quality requirements 2.7 Following good workmanship practices to ensure consistent quality output
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified the jute sector and its main activities
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	1.2 understood basic roles in jute production processes 1.3 recognized common tools and equipment used 1.4 understood simple quality requirements in jute work
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY GREEN PRACTICES IN JUTE SECTOR	Nominal Duration: 10 hrs.	Unit Code: SICIP-JUTE-BPD-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for apply green practices in Jute sector. It specifically includes the tasks of interpreting green concepts, minimizing resources use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 Basic concepts of environmental sustainability and green practices are explained clearly. 1.2 <u>Environmental impacts of jute production</u> processes are identified. 1.3 <u>Resource efficiency concepts</u> are interpreted. 1.4 Waste reduction, reuse, and recycling principles are described in relation to jute operations. 1.5 <u>Eco-friendly practices in the jute sector</u> are recognized and related to workplace activities.
2. Minimize resources use in the workplace	2.1 Water, energy, and raw materials are used efficiently according to workplace requirements. 2.2 <u>Unnecessary consumption of resources</u> is identified and avoided during operations. 2.3 Machines and equipment are operated in energy-efficient modes where applicable. 2.4 Material wastage is minimized through careful handling and accurate measurement. 2.5 <u>Resource-saving practices</u> are applied where possible. 2.6 Leakage, spillage, and wastage of materials are promptly controlled and reported. 2.7 <u>Workplace practices</u> are followed.
3. Implement waste management practices	3.1 <u>Types of waste</u> generated in jute operations are identified correctly. 3.2 Waste materials are segregated according to workplace and environmental requirements. 3.3 <u>Reusable and recyclable materials</u> are separated from non-recyclable waste. 3.4 Waste is stored safely in designated containers or areas. 3.5 Waste disposal is carried out following environmental rules and organizational procedures.

	3.6 Hazardous or chemical waste is handled carefully according to safety guidelines.
	3.7 Waste reduction practices are applied to minimize environmental impact during operations.

Range of Variables

Variable	Range (May include but not limited to):
1. Environmental impacts of jute production	1.1 Water pollution from retting and washing 1.2 Chemical discharge from dyeing and finishing 1.3 Solid waste generation (jute dust, fibre waste) 1.4 Air pollution from boilers and machines 1.5 Energy consumption impacts
2. Resource efficiency concepts	2.1 Water conservation techniques 2.2 Energy-efficient machine operation 2.3 Optimized chemical usage 2.4 Material loss reduction 2.5 Process efficiency improvement
3. Eco-friendly practices in the jute sector	3.1 Use of natural dyes or low-impact dyes 3.2 Biodegradable materials use 3.3 Compliance with environmental regulations 3.4 Proper effluent treatment practices 3.5 Energy-saving production methods
4. Unnecessary consumption of resources	4.1 Overuse of water or chemicals 4.2 Idle machine running 4.3 Excess raw material usage 4.4 Poor process planning or duplication of work
5. Resource-saving practices	5.1 Reuse of rinse water where applicable 5.2 Recycling of jute waste and scraps 5.3 Reusing packaging materials 5.4 Recovery of usable materials from waste
6. Workplace practices	6.1 Following environmental policies and SOPs 6.2 Proper waste segregation and disposal 6.3 Use of eco-friendly materials 6.4 Compliance with environmental regulations 6.5 Maintaining clean and safe work environment
7. Types of waste	7.1 Jute fibre waste and dust 7.2 Yarn and fabric off-cuts 7.3 Chemical waste (dyes, salts, auxiliaries) 7.4 Wastewater from processing 7.5 Packaging waste (poly bags, cartons) 7.6 Sludge from effluent treatment
8. Reusable and recyclable materials	8.1 Jute fibre scraps for reprocessing 8.2 Reusable packaging materials 8.3 Recyclable plastics and paper 8.4 Recoverable chemical containers (if applicable)

Curricular Content Guide

1. Underpinning Knowledge	1.1 Concepts of environmental sustainability, green practices, and eco-friendly workplace behavior. 1.2 Environmental impacts of jute production and related industrial processes. 1.3 Principles of resource efficiency, including water, energy, and raw material conservation. 1.4 Waste reduction strategies such as reuse, recycling, and minimizing material wastage. 1.5 Efficient use of machines and equipment with focus on energy-saving practices. 1.6 Identification, segregation, and safe handling of different types of waste in jute operations. 1.7 Proper waste storage, disposal methods, and compliance with environmental regulations. 1.8 Hazardous waste handling and application of workplace environmental safety procedures.
2. Underpinning Skills	2.1 Applying resource efficiency principles in workplace operations 2.2 Applying waste reduction, reuse, and recycling practices in jute production activities 2.3 Promoting eco-friendly practices in daily workplace tasks 2.4 Using water, energy, and raw materials efficiently according to workplace requirements 2.5 Minimizing unnecessary consumption and wastage of resources during operations 2.6 Operating machines and equipment in energy-efficient modes where applicable 2.7 Handling materials carefully to reduce wastage and improve accuracy 2.8 Controlling leakage, spillage, and resource wastage and reporting issues promptly 2.9 Segregating different types of waste generated in jute operations 2.10 Handling hazardous waste according to safety guidelines and regulations
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource	The following resources must be provided:

Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimized resources use in the workplace 1.3 implemented waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: IDENTIFY MATERIALS, EQUIPMENT AND PROCESSES	Nominal Duration: 20 Hrs.	Unit Code: SICIP-JUTE-BPD-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret raw jute and grades. It specifically includes the task of identifying raw jute and grades, identifying batching oils and chemicals, identifying tools and processing machine, identifying processing machine and parts and functions, describing batching and preparing methods, and describing drawing process.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify raw jute and grades	1.1 Jute is defined as a natural fiber. 1.2 <u>Types of jute</u> are identified based on their source and characteristics. 1.3 <u>Different grades of jute</u> are recognized. 1.4 Sorting procedures of jute bale are identified based on grade and suitability for processing.
2. Identify batching oils and chemicals	2.1 <u>Jute batching oils</u> are identified based on their type and purpose in jute processing. 2.2 <u>Chemicals</u> are identified based on their type and purpose in jute processing. 2.3 Quality and suitability of oils and chemicals are described before use.
3. Identify tools and processing machine	3.1 <u>Common tools</u> are identified based on their type and use in jute processing work. 3.2 <u>Batching machines</u> are identified based on their functions in the production line. 3.3 <u>Preparing machines</u> are recognized. 3.4 <u>Drawing machines</u> are identified based on their functions in the production line. 3.5 Functions and basic operations of each machine are understood according to process requirements.
4. Identify processing machine parts and functions	4.1 <u>Parts of batching machines</u> are identified and their functions are explained in the batching process. 4.2 <u>Components of preparing machines</u> are identified with their roles and functions. 4.3 <u>Parts of drawing machines</u> are identified and their functions. 4.4 Functions of each machine and its components are explained according to process requirements.

5. Describe batching and preparing methods	5.1 Different batching methods are explained based on manual and mechanical application. 5.2 Importance of correct batching proportion and conditioning time is described. 5.3 Preparing method is described using breaker and finisher card machines. 5.4 Stages of preparing are explained including breaking, carding, and leveling of fibers. 5.5 Role of proper batching and preparing in ensuring smooth drawing and better yarn quality is described.
6. Describe drawing process	6.1 Drawing process is described as the stage where jute fibers are drafted and aligned into uniform slivers. 6.2 Slivers from previous stages are fed into drawing frames for further straightening and blending. 6.3 Different stages are explained based on gradual improvement of fiber alignment. 6.4 Drafting zone are used to reduce thickness and improve parallelization of fibers. 6.5 Importance of drawing in preparing fibers for spinning and improving yarn quality is explained.

Range of Variables

Variable	Range (Includes but not limited to):
1. Types of jute	1.1 Tossa 1.2 White 1.3 Mesta 1.4 Kenaf
2. Different grades of jute	Tossa 2.1 KACCHA (Raw): Top, Grade-A, Grade-B, Grade-C, Cross, SMR 2.2 PACCA (Processed): Special, Grade-A, Grade-B, Grade-C, Grade-D, Grade-E. White (Deshi) 2.3 KACCHA (Raw): Top, Grade-A, Grade-B, Grade-C, Cross, SMR 2.4 PACCA (Processed): Special, Grade-A, Grade-B, Grade-C, Grade-D, Grade-E Mesta 2.5 KACCHA (Raw): Top, Grade-A, Grade-B, Grade-C. 2.6 PACCA (Processed): Special, Grade-A, Grade-B, Grade-C. Kenaf 2.7 KACCHA (Raw): Top, Grade-A, Grade-B, Grade-C 2.8 PACCA (Processed): Special, Grade-A, Grade-B, Grade-C
3. Jute batching oils	3.1 Mineral Oil (Batching Oil) → Main oil used for softening and lubrication

	3.2 Vegetable Oil → Eco-friendly alternative in some mills
4. Chemicals	4.1 Emulsifier → Helps oil and water mix properly 4.2 Wetting Agent → Ensures uniform penetration into fibre 4.3 Softener → Improves flexibility and handling 4.4 Antistatic Agent → Reduces static during processing
5. Common tools	5.1 Knife / Cutter 5.2 Hook (Jute Hook) 5.3 Weighing Scale 5.4 Measuring Tape 5.5 Oil Can / Sprayer 5.6 Bucket / Drum 5.7 Brush
6. Batching machine	6.1 Emulsion plant 6.2 Softener 6.3 Spreader
7. Preparing Machine	7.1 Breaker card machine 7.2 Finisher card machine
8. Drawing machine	8.1 First drawing frame 8.2 Second drawing frame 8.3 Third drawing frame/ finisher drawing
9. Parts of batching machines	9.1 Feed hopper 9.2 Feeding conveyor 9.3 Weighing system (scale/batching scale) 9.4 Mixing chamber 9.5 Blending rollers 9.6 Moisture spraying unit 9.7 Control panel 9.8 Discharge conveyor 9.9 Drive motor 9.10 Gear system 9.11 Transmission belt/chain system 9.12 Frame and base structure 9.13 Safety guard system 9.14 Dust control system
10. Components of preparing machines	10.1 Feeding system (hopper/feed conveyor) 10.2 Carding rollers 10.3 Beater rollers 10.4 Drafting rollers 10.5 Cylinder and worker rollers 10.6 Comb or stripper mechanism 10.7 Licker-in roller 10.8 Doffer roller 10.9 Apron or web formation system 10.10 Delivery rollers 10.11 Drive motor 10.12 Gear and transmission system 10.13 Control panel

	10.14 Frame and machine body 10.15 Lubrication system 10.16 Safety guards
11. Parts of drawing machines	11.1 Feeding creel 11.2 Feed rollers 11.3 Drafting rollers 11.4 Drawing rollers (front, middle, back) 11.5 Pressure rollers 11.6 Condenser system 11.7 Sliver guide 11.8 Delivery rollers 11.9 Coiler head 11.10 Can (sliver can) 11.11 Drive motor 11.12 Gear and transmission system 11.13 Control panel 11.14 Frame and machine body 11.15 Lubrication system 11.16 Safety guards

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types, characteristics, grading, and sorting methods of raw jute used in processing. 1.2 Types, properties, and applications of batching oils, emulsions, and processing chemicals. 1.3 Functions and operating principles of batching, preparing, and drawing machines. 1.4 Major parts and components of batching, preparing, and drawing machines and their functions. 1.5 Principles and methods of jute batching, including oil application, moisture conditioning, and batching proportions. 1.6 Fundamentals of preparing processes, including breaking, carding, leveling, and sliver formation. 1.7 Principles of the drawing process, including fiber drafting, blending, alignment, and sliver quality improvement. 1.8 Relationship between batching, preparing, drawing processes, and the production of high-quality jute yarn.
2. Underpinning Skills	2.1. Selecting suitable oils and chemicals based on quality and processing requirements 2.2. Identifying tools, batching, preparing, and drawing machines used in production 2.3. Understanding the functions and basic operation of processing machines and components 2.4. Identifying machine parts and explaining their roles in batching, preparing, and drawing processes

	2.5. Applying knowledge of batching methods, proportions, and conditioning requirements 2.6. Understanding preparing processes including breaking, carding, and fiber leveling 2.7. Recognizing drafting principles to improve fiber parallelization and sliver uniformity 2.8. Applying process knowledge to support improved yarn quality and efficient production
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Raw jute 4.3 Batching oils and chemicals 4.4 Processing machine 4.5 Machine parts 4.6 Tools and equipment

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified raw jute and grades 1.2 identified batching oils and chemicals 1.3 identified tools and processing machine 1.4 identified processing machine parts and functions 1.5 described batching and preparing methods 1.6 described drawing process
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM BATCHING OPERATION	Nominal Duration: 70 Hrs.	Unit Code: SICIP-JUTE-BPD-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform batching operation. It specifically includes the task of selecting and sorting out jute for batching, measuring oil, water and emulsifier ratios, preparing emulsion solutions, operating softener and spreader machine and storing piled jute properly.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Select jute for batching	1.1 Jute is identified according to batch production requirements. 1.2 Grade, type, and condition of jute are checked following workplace procedures. 1.3 Moisture content and cleanliness of jute are inspected. 1.4 <u>Defective, contaminated, or unsuitable jute</u> is separated. 1.5 Selected jute is arranged for batching according to production sequence.
2. Measure oil, water and emulsifier ratios	2.1 <u>Batch formula and mixing ratio</u> are interpreted from production instructions. 2.2 Required quantities of oil, water, and emulsifier are calculated. 2.3 <u>Measuring instruments</u> are selected and checked for accuracy. 2.4 Oil, water, and emulsifier are measured according to specified ratios. 2.5 Measured materials are prepared for emulsion mixing following safety procedures.
3. Prepare emulsion solutions	3.1 <u>Mixing tank and equipment</u> are cleaned and prepared. 3.2 Measured oil, water, and emulsifier are added in the correct sequence. 3.3 Emulsion is mixed until uniform consistency is achieved. 3.4 Concentration and quality of emulsion are checked according to specifications. 3.5 Prepared emulsion is transferred to the batching system for use.
4. Operate softener and spreader machine	4.1 Softener and spreader machine are inspected before operation. 4.2 <u>Machine settings</u> are adjusted according to jute type

	and production requirements. 4.3 Emulsion is applied evenly to jute during processing. 4.4 Machine is operated following standard operating and safety procedures. 4.5 Processed jute is checked for softness and uniform moisture distribution. 4.6 <u>Machine performance issues</u> are identified and reported.
5. Store piled jute properly	5.1 Batched jute is piled according to workplace procedures. 5.2 Piles are covered and labeled with relevant batch information. 5.3 <u>Storage conditions</u> are maintained to preserve moisture and quality. 5.4 Jute is stored for the specified conditioning period. 5.5 Stored jute is monitored to ensure readiness for the next process.

Range of Variables

Variable	Range (Includes but not limited to):
1. Defective, contaminated, or unsuitable jute	1.1 Damaged or weak jute fibers 1.2 Dirty or contaminated jute (sand, oil, waste) 1.3 Wet or rotten jute 1.4 Mixed with foreign materials 1.5 Low-quality or discolored jute 1.6 Not suitable for processing
2. Batch formula and mixing ratio	2.1 Proper proportion of oil, water, and emulsifier 2.2 Measured accurately as per requirement 2.3 Mixed uniformly for processing 2.4 Adjusted based on jute quality and production need
3. Measuring instruments	3.1 Weighing scale 3.2 Measuring cylinder 3.3 Measuring tank 3.4 Flow meter 3.5 Measuring bucket 3.6 Graduated container 3.7 Digital balance
4. Mixing tank and equipment	4.1 Mixing tank 4.2 Agitator (stirrer) 4.3 Pump 4.4 Pipe and valve system 4.5 Motor and drive unit 4.6 Measuring instruments 4.7 Drainage system
5. Machine settings	5.1 Feed speed setting 5.2 Roller pressure setting

	5.3 Oil/water spray rate adjustment 5.4 Moisture level setting 5.5 Machine speed control 5.6 Temperature setting (if applicable) 5.7 Spreading width adjustment 5.8 Timing/control panel setting 5.9 Tension adjustment 5.10 Safety setting/checks
6. Machine performance issues	6.1 Uneven softening 6.2 Irregular spreading 6.3 Low machine speed 6.4 Blockage in feed system 6.5 Roller slippage 6.6 Excess or insufficient moisture 6.7 Mechanical vibration or noise 6.8 Motor or power failure 6.9 Poor lubrication 6.10 Inconsistent output quality
7. Storage conditions	7.1 Dry and well-ventilated area 7.2 Free from moisture and rain exposure 7.3 Clean storage space without dirt or pests 7.4 Proper stacking to allow airflow 7.5 Avoid direct sunlight 7.6 Safe distance from chemicals or fire sources 7.7 Raised platform or pallets used 7.8 Regular inspection of stored jute

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of jute selection, grading, moisture assessment, and quality inspection for batching. 1.2 Batch formulation concepts, including oil, water, and emulsifier ratios used in jute processing. 1.3 Methods of calculating, measuring, and preparing batching emulsion according to production specifications. 1.4 Functions and operating principles of softener and spreader machines in the batching process. 1.5 Techniques for applying emulsion uniformly to achieve required softness and moisture distribution. 1.6 Standard operating procedures (SOPs) and safety practices for batching operations and machine use. 1.7 Methods for piling, labeling, storing, and conditioning batched jute to maintain quality. 1.8 Quality control requirements, machine performance monitoring, and preparation of batched jute for subsequent processing stages.
2. Underpinning Skills	2.1 Inspecting jute quality and separating defective or

	contaminated materials 2.2 Applying safe procedures for measuring and handling batching chemicals 2.3 Preparing uniform emulsion solutions following correct mixing sequences 2.4 Checking emulsion concentration and quality against production specifications 2.5 Operating softener and spreader machines according to standard operating procedures 2.6 Adjusting machine settings based on jute type and production requirements 2.7 Applying emulsion evenly to achieve uniform softness and moisture distribution 2.8 Labeling, monitoring, and preparing stored jute for the next production process
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Raw jute 4.3 Oil 4.4 Emulsion solutions 4.5 Softener and spreader machine

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 selected and sorted out jute for batching 1.2 measured oil, water and emulsifier ratios 1.3 prepared emulsion solutions 1.4 operated softener and spreader machine 1.5 stored piled jute properly
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or

	simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: PERFORM JUTE PREPARING OPERATIONS	Nominal Duration: 40 Hrs.	Unit Code: SICIP-JUTE-BPD-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform jute preparing operations. It specifically includes the task of feeding piled jute into preparing machines, adjusting machine settings and controls, operating breaker card machine, monitoring fibre condition and output and maintaining continuous material flow.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Feed piled jute into preparing machines	1.1 Batched and conditioned jute is identified according to production requirements. 1.2 Piled jute is transported to preparing machines following workplace procedures. 1.3 Jute is opened and separated to ensure smooth feeding. 1.4 Jute is fed uniformly into the breaker card machine. 1.5 Feeding is carried out in accordance with machine capacity and safety requirements.
2. Adjust machine settings and controls	2.1 Production specifications and <u>machine operating parameters</u> are reviewed. 2.2 <u>Machine settings</u> are adjusted according to jute grade and processing requirements. 2.3 <u>Safety guards and control systems</u> are checked before operation. 2.4 Adjustments are verified to ensure proper machine performance.
3. Operate breaker card and finisher card machine	3.1 Breaker card machine is started and operated following standard procedures. 3.2 Material output from the breaker card is transferred to the finisher card machine. 3.3 Finisher card machine is operated to further straighten and clean fibres. 3.4 Machine performance is monitored throughout operation. 3.5 <u>Operational abnormalities</u> are identified and reported according to workplace procedures.
4. Monitor fibre condition and output	4.1 Fibre sliver is checked for cleanliness, alignment, and uniformity. 4.2 <u>Fibre defects</u> are identified. 4.3 Output quality is compared with production specifications. 4.4 Corrective adjustments are made to maintain required

	quality. 4.5 Non-conforming material is isolated and reported.
5. Maintain continuous material flow	5.1 Material flow between machines is monitored continuously. 5.2 <u>Sliver cans and collection systems</u> are positioned and replaced as required. 5.3 Feed interruptions and blockages are cleared promptly and safely. 5.4 Machine stoppages are minimized through timely intervention. 5.5 Production continuity is maintained according to operational requirements.

Range of Variables

Variable	Range (Includes but not limited to):
1. Machine operating parameters	1.1 Machine speed (RPM) 1.2 Feed rate 1.3 Roller pressure 1.4 Moisture level 1.5 Draft ratio 1.6 Temperature (if applicable) 1.7 Fiber flow rate 1.8 Spreading width 1.9 Lubrication level 1.10 Power/load setting
2. Machine settings	2.1 Feed speed adjustment 2.2 Roller pressure setting 2.3 Drafting ratio setting 2.4 Moisture control setting 2.5 Machine speed control 2.6 Spreading width adjustment 2.7 Tension adjustment 2.8 Lubrication setting 2.9 Alignment of rollers 2.10 Safety guard and control settings
3. Safety guards and control systems	3.1 Protective safety covers (guards) 3.2 Emergency stop button 3.3 Safety interlock system 3.4 Control panel 3.5 Sensor-based safety system 3.6 Machine start/stop controls 3.7 Overload protection system 3.8 Warning alarms or indicators
4. Operational abnormalities	4.1 Uneven sliver formation 4.2 Fiber wrapping on rollers 4.3 Machine vibration or noise

	4.4 Sliver breakage or blockage 4.5 Irregular feed of material 4.6 Roller slippage 4.7 Overheating of machine parts 4.8 Low output or reduced production rate 4.9 Jammed or clogged feed system 4.10 Poor carding quality output
5. Fibre defects	5.1 Short fibre length 5.2 Weak or damaged fibres 5.3 Knotted or entangled fibres 5.4 Dirty or contaminated fibres 5.5 Uneven fibre thickness 5.6 Discoloured fibres 5.7 Coarse or rough fibres 5.8 Mixed foreign materials 5.9 Moisture-affected fibres 5.10 Poor quality fibre output
6. Sliver cans and collection systems	6.1 Sliver cans 6.2 Can coiler system 6.3 Coiling mechanism 6.4 Sliver guide tubes 6.5 Delivery rollers 6.6 Collection conveyor system 6.7 Automatic can changer 6.8 Sliver transport system 6.9 Can lifting and positioning unit 6.10 Storage racks for filled cans

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of preparing processes, including fibre opening, cleaning, carding, and sliver formation. 1.2 Functions, components, and operating principles of breaker card and finisher card machines. 1.3 Machine setting parameters based on jute grade, fibre characteristics, and production requirements. 1.4 Standard operating procedures (SOPs) and workplace safety requirements for preparing machines. 1.5 Methods for monitoring fibre quality, including cleanliness, alignment, uniformity, and defect identification. 1.6 Techniques for maintaining continuous material flow and efficient sliver handling during production. 1.7 Procedures for identifying operational abnormalities, blockages, and machine performance issues. 1.8 Quality control requirements and corrective actions to ensure preparing process efficiency and consistent sliver quality.
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2. Underpinning Skills	2.1. Applying safe and uniform feeding techniques for efficient machine operation 2.2. Adjusting machine settings based on jute grade and processing specifications 2.3. Inspecting safety guards and control systems before machine operation 2.4. Operating breaker card and finisher card machines following standard procedures 2.5. Monitoring machine performance and identifying operational abnormalities 2.6. Assessing fibre sliver quality for cleanliness, alignment, and uniformity 2.7. Applying corrective adjustments to maintain required fibre quality 2.8. Isolating and reporting non-conforming materials according to workplace procedures 2.9. Maintaining continuous material flow between processing machines
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Preparing machine 4.3 Breaker and finisher card machine 4.4 Standard operating procedure (samples)

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 feed piled jute into preparing machines 1.2 adjusted machine settings and controls 1.3 operated breaker card and finisher card machine 1.4 monitored fibre condition and output 1.5 maintained continuous material flow
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)

3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: PERFORM DRAWING OPERATIONS	Nominal Duration: 30 Hrs.	Unit Code: SICIP-JUTE-BPD-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform drawings operations. It specifically includes the task of feeding slivers into drawing machines, adjusted draft and speed settings, operated drawing frames, monitored slivers quality and uniformity and handled drawing output materials properly.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Feed slivers into drawing machines	1.1 Finisher carded slivers are identified according to <u>production requirements</u> . 1.2 Sliver cans are positioned correctly at the drawing frame. 1.3 <u>Sliver ends</u> are inspected and joined as required. 1.4 Multiple slivers (Doubling) are fed into the drawing machine in the correct sequence. 1.5 Feeding is carried out uniformly to ensure continuous processing.
2. Adjust draft and speed settings	2.1 Production specifications and <u>machine parameters</u> are reviewed. 2.2 <u>Draft ratio</u> is set according to required sliver count and uniformity. 2.3 Delivery speed is adjusted to meet production requirements. 2.4 Roller pressure and tension settings are adjusted as specified. 2.5 Settings are checked and verified before operation.
3. Operate drawing frames	3.1 Drawing frame is started following <u>standard operating procedures</u> . 3.2 Slivers are drawn and combined to improve fibre parallelization and evenness. 3.3 Machine controls are operated in accordance with production requirements. 3.4 Machine performance is monitored continuously during operation. 3.5 <u>Operational faults</u> and abnormal conditions are identified and reported.
4. Monitor sliver quality and uniformity	4.1 Drawn sliver is checked for evenness, strength, and cleanliness. 4.2 <u>Variations in sliver thickness and defects</u> are identified. 4.3 Output quality is compared with production

	specifications. 4.4 Machine settings are adjusted to correct quality deviations. 4.5 Non-conforming sliver is isolated and reported according to workplace procedures.
5. Handle drawing output materials properly	5.1 Drawn sliver is collected in cans or containers without damage. 5.2 Filled cans are labeled with relevant production information. 5.3 Output materials are transferred and stored according to workplace procedures. 5.4 Sliver is protected from contamination and deformation during handling. 5.5 Work area is maintained clean and organized after completion of operations.

Range of Variables

Variable	Range (Includes but not limited to):
1. Production requirements	1.1 Production order 1.2 Yarn count specification 1.3 Quality standard 1.4 Machine capacity 1.5 Process schedule
2. Sliver ends	2.1 Intact 2.2 Broken 2.3 Uneven 2.4 Tangled
3. Machine parameters	3.1 Draft ratio 3.2 Delivery speed 3.3 Roller pressure 3.4 Tension settings 3.5 Roller settings
4. Draft ratio	4.1 Sliver count 4.2 Fiber characteristics 4.3 Product requirements
5. Standard operating procedures	5.1 Workplace operating procedures 5.2 Manufacturer's requirements 5.3 Production guidelines 5.4 Safety procedures
6. Operational faults	6.1 Sliver breakage 6.2 Under/over moisturized sliver 6.3 Roller lapping 6.4 Uneven sliver 6.5 Uneven joints 6.6 Machine stoppage 6.7 Excessive vibration

	6.8 Mechanical malfunction
7. Variations in sliver thickness and defects	7.1 Thick places 7.2 Thin places 7.3 Uneven weight distribution 7.4 Irregular sliver formation Defects 7.5 Entanglements 7.6 Fibre lumps 7.7 Broken slivers 7.8 Uneven blending 7.9 Over riding
8. Output materials	8.1 Drawn slivers 8.2 Filled cans (Packing) 8.3 Rejected materials 8.4 Production outputs

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of the drawing process, including doubling, drafting, fibre parallelization, and sliver equalization. 1.2 Functions, components, and operating principles of drawing frames and related equipment. 1.3 Machine setting parameters, including draft ratio, delivery speed, roller pressure, and tension control. 1.4 Standard operating procedures (SOPs) and workplace safety requirements for drawing operations. 1.5 Methods for monitoring sliver quality, including evenness, strength, cleanliness, and defect identification. 1.6 Techniques for identifying operational faults and making machine adjustments to maintain product quality. 1.7 Procedures for collecting, labeling, handling, storing, and protecting drawn slivers from damage or contamination. 1.8 Quality control practices and production documentation to ensure consistent sliver quality and efficient drawing operations.
2. Underpinning Skills	2.1. Positioning sliver cans and joining sliver ends correctly for continuous processing 2.2. Applying doubling techniques to improve fibre blending and uniformity 2.3. Adjusting draft ratio, delivery speed, roller pressure, and tension according to specifications 2.4. Verifying machine settings before starting drawing operations 2.5. Operating drawing frames following standard operating procedures

	2.6. Identifying sliver defects and comparing output with production specifications 2.7. Adjusting machine settings to maintain required sliver quality 2.8. Maintaining a clean and organized work area after drawing operations
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Drawing machine

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 feeded slivers into drawing machines 1.2 adjusted draft and speed settings 1.3 operated drawing frames 1.4 monitored sliver quality and uniformity 1.5 handled drawing output materials properly
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MAINTAIN EQUIPMENT AND WORK AREA	Nominal Duration: 20 Hrs.	Unit Code: SICIP-JUTE-BPD-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to maintain equipment and work area. It specifically includes the task of performing cleaning of machines and work areas, conducting routine equipment checking, carrying out basic machine maintenance & maintaining storing tools and equipment properly.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Perform cleaning of machines and work areas	1.1 Machines are stopped and isolated according to standard safety procedures before cleaning. 1.2 Dust, fibre waste, oil, and other contaminants are removed from machines and surrounding areas. 1.3 <u>Machine parts</u> , guards, and accessible components are cleaned. 1.4 Floors, walkways, and workstations are cleaned and kept free from obstructions. 1.5 <u>Waste materials</u> are collected and disposed of according to workplace procedures.
2. Conduct routine equipment checking	2.1 Machines and equipment are checked for visible signs of wear, damage, and malfunction. 2.2 Belts, rollers, bearings, guards, and control devices are checked for proper condition. 2.3 Unusual noise, vibration, leakage, or overheating is identified during checking. 2.4 <u>Safety devices</u> and emergency controls are tested for proper operation. 2.5 <u>Faults and abnormal conditions</u> are recorded according to workplace procedures.
3. Carry out basic machine maintenance	3.1 Routine maintenance requirements are identified according to machine manuals. 3.2 <u>Machine components</u> are checked for proper condition. 3.3 Lubrication is carried out using specified lubricants and maintenance schedules. 3.4 Loose parts, fittings, and fasteners are checked and tightened as required. 3.5 <u>Basic maintenance activities</u> are conducted following safety procedures and lockout requirements.
4. Maintain and store tools and equipment	4.1 Tools and equipment are cleaned after use according to workplace procedures.

properly	4.2 Damaged or defective tools and equipment are identified and reported to appropriate personnel. 4.3 Tools and equipment are stored safely in designated locations after use. 4.4 Storage conditions are maintained to prevent damage, corrosion, or contamination.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Machine parts	1.1 Frames 1.2 Covers 1.3 Panels 1.4 Rollers 1.5 Feed side 1.6 Delivery side 1.7 External machine parts
2. Waste materials	2.1 Dust 2.2 Fibre waste 2.3 Used cleaning materials 2.4 Oil-contaminated waste 2.5 General production waste
3. Safety devices	3.1 Emergency Stop Button 3.2 Roller Guard 3.3 Belt and Pulley Guard 3.4 Gear Guard 3.5 Overload Protection 3.6 Safety Covers 3.7 Warning Signs
4. Faults and abnormal conditions	4.1 Mechanical faults 4.2 Electrical faults 4.3 Safety defects 4.4 Operational irregularities 4.5 Performance deviations
5. Machine components	5.1 Rollers 5.2 Gears 5.3 Belts 5.4 Chains 5.5 Pulleys 5.6 Shafts 5.7 Motors 5.8 Lubrication points 5.9 Guards 5.10 Fasteners
6. Basic maintenance activities	6.1 Cleaning 6.2 Lubrication 6.3 Tightening

	6.4 Minor adjustments
	6.5 Component inspection
	6.6 Replacement of minor consumables

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of workplace housekeeping and the importance of maintaining clean and safe production areas. 1.2 Standard cleaning procedures for jute processing machines, equipment, and workstations. 1.3 Basic machine components, their functions, and common signs of wear, damage, or malfunction. 1.4 Routine inspection techniques for belts, rollers, bearings, guards, and safety devices. 1.5 Principles of preventive maintenance, lubrication, and tightening of machine components. 1.6 Lockout/tagout (LOTO), machine isolation, and safe maintenance procedures. 1.7 Proper handling, maintenance, storage, and protection of tools and equipment. 1.8 Procedures for fault reporting, maintenance record keeping, and safe disposal of waste materials.
2. Underpinning Skills	2.1 Cleaning machines, work areas, and components to maintain safe operating conditions 2.2 Removing dust, fibre waste, oil, and contaminants according to workplace procedures 2.3 Disposing of waste materials following workplace and environmental requirements 2.4 Checking belts, rollers, bearings, guards, and control devices for proper condition 2.5 Recording faults and abnormal conditions according to workplace procedures 2.6 Performing routine machine maintenance, including lubrication and tightening of components 2.7 Following lockout and safety procedures during maintenance activities 2.8 Cleaning, inspecting, and storing tools and equipment properly after use
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors

	in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Tools and equipment 4.3 Cleaning materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 performed cleaning of machines and work areas 1.2 conducted routine equipment checking 1.3 carried out basic machine maintenance 1.4 maintained and stored tools and equipment properly
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MAINTAIN QUALITY AND PRODUCTION EFFICIENCY	Nominal Duration: 20 Hrs.	Unit Code: SICIP-JUTE-BPD-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to maintain quality and production efficiency. It specifically includes the task of checking product quality standards, identifying defects and irregularities, taking corrective actions during production, maintaining production targets and efficiency and following workplace rules and procedures.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Check product quality standards	1.1 Product quality requirements are identified according to standards and production instructions. 1.2 Jute materials and slivers are visually checked for cleanliness, uniformity, and appearance. 1.3 Product output is compared with specified quality requirements. 1.4 Non-conforming materials are separated and reported to the supervisor.
2. Identify defects and irregularities	2.1 <u>Common defects and irregularities</u> are identified in jute materials and production processes. 2.2 Variations in sliver thickness, fibre alignment and moisture contents are checked against specifications. 2.3 Defective or non-conforming materials are separated according to workplace procedures. 2.4 Causes of visible defects and irregularities are identified and reported to the supervisor. 2.5 <u>Corrective actions</u> within the operator's responsibility are taken to minimize quality deviations.
3. Take corrective actions during production	3.1 <u>Quality deviations</u> and production issues are identified during routine operations. 3.2 Minor machine settings are adjusted according to workplace procedures to maintain product quality. 3.3 Defective materials are removed or separated to prevent further quality issues. 3.4 Operational abnormalities beyond the operator's authority are reported to the supervisor promptly.
4. Maintain production targets and efficiency	4.1 Daily production targets and work instructions are identified and followed. 4.2 Machines are operated according to standard procedures to achieve required output levels. 4.3 Material flow is maintained to minimize interruptions and

	production delays. 4.4 <u>Wastage of raw materials</u> and processing materials is minimized during operations. 4.5 Downtime and production losses are reported promptly to the supervisor.
5. Follow workplace rules and procedures	5.1 Workplace policies, standard operating procedures, and safety requirements are followed. 5.2 Personal protective equipment is used according to organizational guidelines. 5.3 Housekeeping standards are maintained in the work area. 5.4 Communication with supervisors and team members is carried out effectively. 5.5 Work activities are performed in compliance with organizational and regulatory requirements.

Range of Variables

Variable	Range (Includes but not limited to):
1. Common defects and irregularities	1.1 Uneven batch 1.2 Improper emulsion application 1.3 Excessive or insufficient moisture 1.4 Fibre entanglement 1.5 Sliver irregularities 1.6 Sliver breakage 1.7 Contamination by dust or foreign materials 1.8 Poor fibre alignment 1.9 Machine feeding irregularities
2. Corrective actions	2.1 Adjusting machine settings 2.2 Correcting material feeding 2.3 Removing defective materials 2.4 Cleaning machine parts 2.5 Clearing blockages 2.6 Reporting faults to supervisor 2.7 Rechecking process parameters
3. Quality deviations	3.1 Variations in sliver weight and strength 3.2 Irregular sliver 3.3 Improper moisture content 3.4 Poor fibre parallelization 3.5 Contamination 3.6 Damaged fibres 3.7 Inconsistent output quality 3.8 Non-compliance with production specifications
4. Wastage of raw materials	4.1 Excessive jute loss during feeding 4.2 Fibre waste from machine droppages 4.3 Damaged slivers 4.4 Spillage of batching emulsion

	4.5	Improper handling of materials
	4.6	Rejected production output
	4.7	Unnecessary trimming and cleaning waste

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of quality control, product specifications, and quality standards in jute processing. 1.2 Common defects in jute materials and slivers, including their causes and identification methods. 1.3 Methods for inspecting sliver quality, fibre alignment, moisture content, cleanliness, and uniformity. 1.4 Basic corrective actions and machine adjustments to maintain product quality during production. 1.5 Production targets, workflow management, and techniques for minimizing downtime and material wastage. 1.6 Standard operating procedures (SOPs), workplace rules, and safe operating practices. 1.7 Effective communication and reporting procedures for quality issues, production problems, and operational abnormalities. 1.8 Importance of housekeeping, PPE, and compliance with organizational and regulatory requirements for efficient production.
2. Underpinning Skills	2.1 Applying product quality standards during jute processing operations 2.2 Inspecting jute materials and slivers for cleanliness, uniformity, and quality 2.3 Comparing product output with specified quality requirements 2.4 Separating and reporting non-conforming materials according to workplace procedures 2.5 Adjusting minor machine settings to maintain product quality and consistency 2.6 Reporting operational abnormalities and quality issues to supervisors promptly 2.7 Following production targets and standard operating procedures to improve efficiency 2.8 Following workplace safety rules, housekeeping standards, and organizational procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace

	3.8	Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1	Workplace (simulated or actual)
	4.2	Quality standards
	4.3	Sample

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 checked product quality standards 1.2 identified defects and irregularities 1.3 take corrective actions during production 1.4 maintained production targets and efficiency 1.5 followed workplace rules and procedures
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Batching, Preparing and Drawing
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptops	1
2.	Multimedia projector & screen	1
3.	Scanner	1
4.	Printer	1
5.	Moister meter	5
6.	Digital Weighing scale	1
7.	Tensile strength tester	5
8.	Measuring tape	15
9.	Calculator	5
10.	Techo meter	2
11.	Hackling machine	5
12.	Emulsion tank	1
13.	Softener machine	1
14.	Spreader machine	1
15.	Breaker card (Hand feed, Roll feed)	2
16.	Finisher card	4
17.	Drawing (first, second and third)	3
18.	Trolly	5
19.	Dollop weight	2
20.	Jute table	2
21.	Pile cover	5
22.	Reserve tank	1
23.	Danage	10
24.	Digital scale (40 kg)	1

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary

training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on **Batching, Preparing and Drawing** the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.